

Question number	Answer	Additional guidance	Mark					
3(a)(i)	<table><tr><td>tryptophan</td><td>serine</td><td>serine</td><td>lysine</td><td>tryptophan</td></tr></table>	tryptophan	serine	serine	lysine	tryptophan	ACCEPT trp for tryptophan ser for serine lys for lysine	(1)
tryptophan	serine	serine	lysine	tryptophan				

Question number	Answer	Additional guidance	Mark
3(a)(ii)	An explanation that includes the following points: - because {three bases form one code (for one amino acid) / there are 5 codons} (1) - because {there is no stop codon in this sequence / it is a non-overlapping (genetic) code} (1)	ACCEPT triplet {codon / code} IGNORE three bases make an amino acid DO NOT ACCEPT degenerate	(2)

Question number	Answer	Additional guidance	Mark
3(b)	An explanation that includes three of the following points: - because it is a degenerate code (1) - which means that there are more codes than {needed / number of amino acids} (1) (arranging the four bases in triplets gives) 64 possible combinations and there are 20 amino acids to code for (and stop codons) (1) - minimises the effect of mutations (1)	DO NOT ACCEPT non-overlapping ALLOW from a description	(3)

Question number	Answer	Additional guidance	Mark
3(c)(i)	95.31 (1)		(1)

Question number	Answer	Additional guidance	Mark
3(c)(ii)	An explanation that includes the following points: (remaining codes are) stop {codons / codes} (on RNA / DNA) (1) - therefore no more amino acids can be added to the {polypeptide chain / protein} (1)	DO NOT ACCEPT start codons ACCEPT ends translation / signals end of (m)RNA	(2)

Question number	Answer	Mark		
3(d)	The only correct answer is B. <table><tr><td>ACC</td><td>UGG</td></tr></table> <i>A is incorrect because T binds to A and C to G on mRNA and U binds to T and G binds to C on tRNA</i> <i>C is incorrect because T binds to A and C to G on mRNA and U binds to T and G binds to C on tRNA</i> <i>D is incorrect because T binds to A and C to G on mRNA and U binds to T and G binds to C on tRNA</i>	ACC	UGG	(1)
ACC	UGG			